Battery Level Indicator

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Objectives

The objective of the Battery Level Indicator is to display the voltage level of a battery shown through the LEDs and shown in message on a LCD screen.

Problem Description

Need Statement:

In this world we live in today, most of everything we use contains some sort of battery. The issue that often arrives is knowing how much charge is still in that battery. Not knowing how to use a complicated device like a multimeter or other tools can lead to device failures or even a power loss you'll never know it can happen. The fundamental need I'm trying to satisfy is having a battery level indicator which fits in our needs of being simple, cost effective and user friendly when it comes to monitoring the battery level. This battery level indicator is going to be helpful for many people using it because of the visual representation of the battery charge status it has. The project we are constructing serves as both a technical solution and consumer product which focuses to attract users to use this product instead of the other complex measuring instruments out there. Also for educational purposes, this product would contain important concepts within like voltage dividers, comparators, and analog indicators.

Problem Definition:

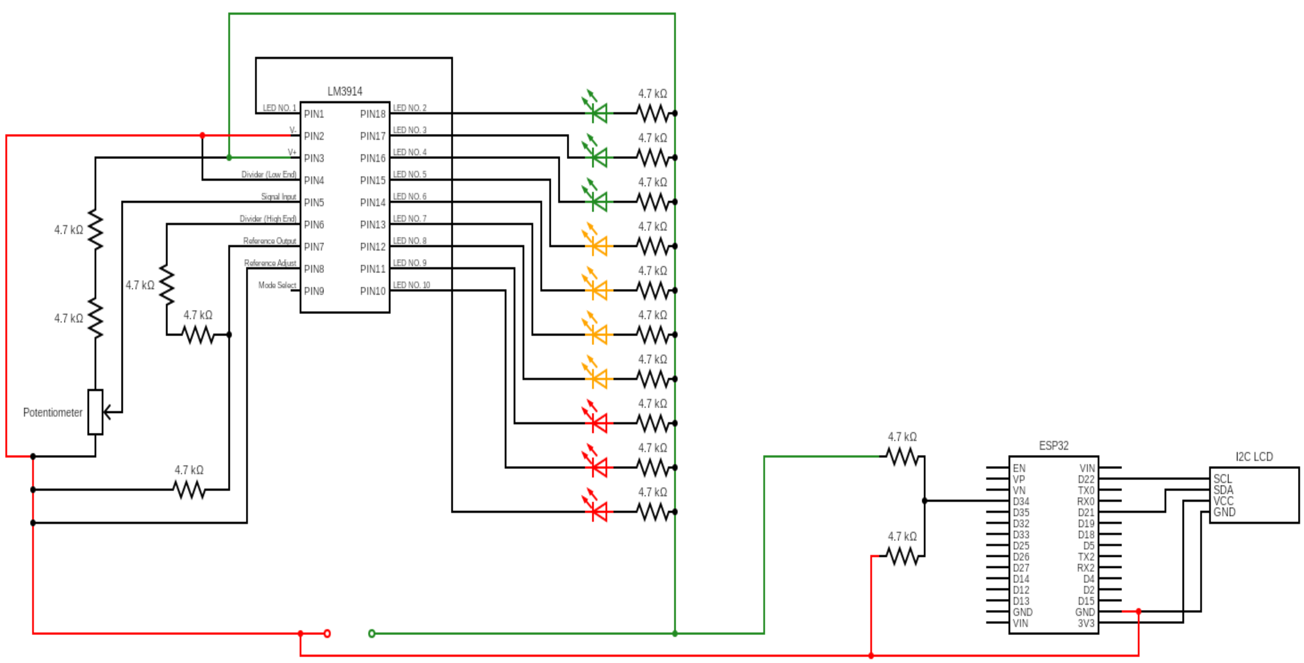
Every one of us in today's age uses some sort of electronics in our day-to-day life that have batteries. The issue is that we never know when that item that uses batteries will die and having a tool like a multimeter might be hard to use to check battery life and cause other issues if we don't have the knowledge. Our project aim is constructing a Battery Level Indicator that uses LED's, voltage divider, and operational amplifiers to represent different voltage levels of a battery we are going to test. The circuit we are creating is going to be fairly easy to put together, its going to be affordable when getting the materials and it will be applicable to many type of batteries. Lastly, the way our product is constructed and the materials that are being used, anyone using this wont have a problem in figuring out how to monitor the health of the battery.

Functional Requirements

1. The built system must be able to detect and display the battery voltage on the LCD
2. The built system must turn all 10 LEDs which correlates to the battery level
3. The voltage reading on the LCD screen must be accurate within 0.2 V
4. The built system has to operate safely from 3 V to 9 V input power

Detail Design Description

The build of our battery level indicator is a simple resistor diode circuit with resistors in series with 10 LEDs all connected to power and a LM3914 chip. Power is also supplied to a voltage divider that then connects to a ESP32 chip that powers and connects to an LCD I2 chip. From the different pin out puts on the LM3914, they're two more voltage dividers and a potentiometer connected to the LM3914. PlatformIO was used to code our ESP32 to display Voltage on the LCD screen, using the C++ coding language.



```cpp
#include <Arduino.h>
#include <LiquidCrystal_I2C.h>

// LCD setup — typical I2C address is 0x27, 16 columns, 2 rows
LiquidCrystal_I2C lcd(0x27, 16, 2);

// Analog pin for voltage divider
const int analogPin = 34;

// Resistor values used in the voltage divider
const float R1 = 22000.0; // 22kΩ
const float R2 = 10000.0;  // 10kΩ

void setup() {
  Serial.begin(9600);
  lcd.init();              // Initialize the LCD
  lcd.backlight();         // Turn on backlight
}

void loop() {
  // Read analog value
  int analogValue = analogRead(analogPin);

  // Print the raw analog value to the Serial Monitor
  Serial.print("Analog Value: ");
  Serial.println(analogValue);  // Prints value between 0 and 4095

  // Convert analog reading to voltage (ESP32 ADC range is 0-4095, 0-3.3V)
  float vOut = (analogValue / 4095.0) * 3.3;

  // Recalculate actual battery voltage using voltage divider formula
  float batteryVoltage = vOut * (R1 + R2) / R2;

  // Print calculated battery voltage to serial for debugging
  Serial.print("Battery Voltage: ");
  Serial.println(batteryVoltage);

  // Display on LCD
  lcd.setCursor(0, 0);
  lcd.print("Battery:");
  lcd.setCursor(0, 1);
  lcd.print(batteryVoltage, 2); // Print with 2 decimal places
  lcd.print(" V");

  delay(1000); // Update every second
}
```

Design Verification

Functional Requirements	**Objective**	**Tests**	**Analyze**
The built system must be able to detect and display the battery voltage on the LCD	The user must be able see the voltage	Added power to ESP32 and uploaded code and measured a voltage and see what voltage was displayed on LCD	Pass
The built system must turn all 10 LEDs which correlates to the battery level	Shows the user visually through lighting if the battery has charge or not. Three colors coordinate with different voltage, green meaning 100-80%, Yellow 70-40%, Red 30-0%. Lights help user tell if battery needs to be replaced.	Applied power with DC Power Supply and measured and tested different voltages ranging from 9V-0V	Pass
The voltage reading on the LCD screen must be accurate within 0.2 V	The voltage must have margin error +-0.2V	Tested our battery by measuring with the Multimeter and then compared voltage measured with voltage displayed on LCD screen	Pass
The built system has to operate safely from 3 V to 9 V input power	Allow the user to be able to test a double A battery and a 9V battery.	Implemented a potentiometer to change the voltage being measured and tested with the Multimeter making sure when changed to 3V and 3V was supplied that the 10th LED would light up and vice versa with 9V.	Pass

1. **Conclusion**

The main objective of this project was to make a Battery level Indicator that would display the voltage level of a battery shown through the LEDs and shown in message on a LCD screen. This project used a multitude of materials starting with a LM3914, an ESP32 microcontroller, 10 LEDs, resistors, potentiometer, LCD screen and wires and breadboard. Also when putting everything together an important thing I made sure that I used a voltage divider so it can scale the battery voltage down to the right level so the ESP32 can read it through the ADC input safely. Then as we were getting ready for the testing process of the project, we made sure that the resistors we picked in series set up the lowest voltage it can read and the maximum voltage. In this case it was 3 V to 9 V which we tested with many voltage sources like a DC Power Supply, AA battery and a 9 V Alkine battery. In this process we looked to see the LM3914 activate its LED bar display with 10 LEDs connected to display how it lights up based on the input voltage. Also we looked at the LCD screen which was connected to the ESP32 to see how it displayed the battery voltage within 0.2 V. From this project I can say 2 of the biggest key challenges was calibrating the voltage divider which connected to the ADC pin of the ESP32 for a accurate reading on the battery and setting up the LED reference voltage range which both were tackled by making multiple adjustments and a lot of testing.

6. **References**

Battery Level Indicator Circuit Principle, Example, Diagram

In-Depth: How to Use an I2C LCD Display With ESP32